

Discussion Unit:

DSE Core (and more), Security

Introduce DSE Security

- Large topic, necessary for the enterprise
- CSO or similar ? Get help
- Multiple Levels
 - OS (Pre DSE)
 - DSE
 - Database connections (end user)
 - Inter-nodal communication
 - Data in flight
 - Data at rest
 - Files, (config, temp)
 - (Other)

Discussion Lab:

What is ? (Define the term)



Source: <https://www.youtube.com/watch?v=5NhZXDU9FMo>

Define the term: What is ?



Authentication
Authorization
Separation of Roles
Encryption at rest
Encryption in flight
RBAC
RLAC
TLAC
Auditing

**“The forecast average loss
for a breach of 1,000 records
is generally over \$\$ (2015)”**

Source: <https://www.verizonenterprise.com/verizon-insights-lab/dbir/>



End of Discussion Lab:

DSE 6 Security: *Overview*

Feature	Core	Search	Analytics	Graph
Authenticate, LDAP or Internal	Yes	Yes	Yes	Partial
Authenticate, Kerberos (Extern)	Yes	Yes	Yes	Yes
Authorize (RBAC)	Yes	Partial	Partial	Yes
Row Level perm (RLAC)	Yes	No	Yes	No
Client-to-node encryption	Yes	Yes	Yes	Yes
Node-to-node encryption	Yes	Yes	Yes	Yes
Transparent data encryption	Yes	Yes	No	Yes
Data auditing	Yes	Yes	Partial	Yes

DSE Security: Authentication

Inter-nodal (gossip, other)

- SSL certificates

"DSE Unified Authentication",
database
connections only



Database connections (client to node, tools and apps)

- [Internal, credentials by DSE](#)
- LDAP (including Active Directory)
- Kerberos



Practice Lab to follow

PKI (Public key infrastructure)

KMIP (Key management interoperability protocol)

- And encrypt data in flight (client to node SSL)

DSE Security: Authorization



Unbreakable

Source: <https://www.imdb.com/title/tt0217869>

Role based access control (RBAC)
-- Only if authentication is enabled

Internal (credentials by DSE)
-- 1:1 mapping of user name to roles

LDAP
-- 1:M mapping, users assigned roles that match groups in LDAP

DSE: Encryption of Data in Flight

SSL encryption for data in-flight for the following components:

- DSE transactional nodes
- DSE Search (Apache Solr™)
- DSE Analytics (Apache Spark™)
- DSE Graph
- DSE tools
- DSE drivers
- DSE OpsCenter



DSE: Encryption of Data at Rest



"Transparent Data Encryption (TDS)"

- Entire tables (except for partition keys which are always stored in plain text)
- SSTables containing data, including system tables (such as `system.batchlog` and `system.paxos`)
- Search indexes
- File-based Hints (in DSE 5.0 and later)
- Commit logs
- Sensitive properties in `dse.yaml` and `cassandra.yaml`

- TDE only applies to data stored in the database. DSE does not support encrypting data that is used by Spark and stored in DSEFS or local temporary directories.
- Graph: Cached data is not encrypted. Encryption may slightly impact performance.

DSE: Securing Ports, Temp directories (JNA)

Port	Service
Public facing ports	
22	SSH (default)
DataStax Enterprise public ports	
(random)	Spark port for the driver to listen on. Used for communicating with the executors and the standalone Master.
(random)	Spark port for all block managers to listen on. These ports exist on both the driver and the executors.
(local hostname)	Spark port for communicating with the executors and the standalone Master.
4040	Spark application web site port. If an application is already using the designated port, DSE tries 4041 and continues incrementing until it finds an open port.
5598, 5599	Public/internode ports for DSE File System (DSEFS) clients.



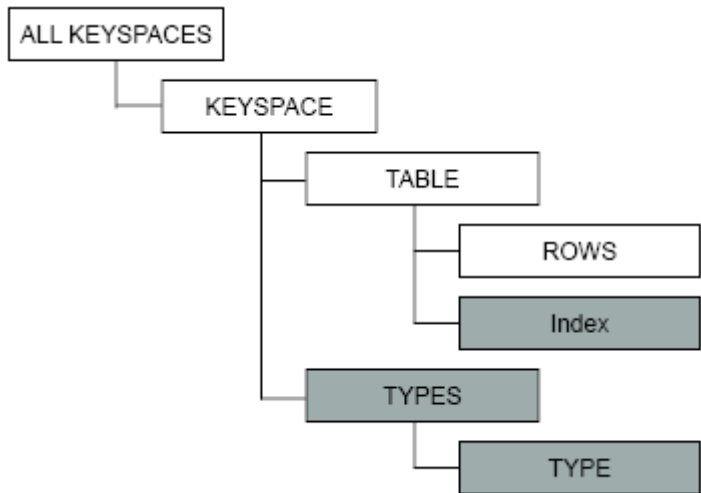
(20+ more)

DSE: RBAC (Managing Roles)

- Similar to any RDBMS
- Keyspace Filtering
cassandra.yaml
system_keyspaces_filtering: true
User/role needs
DESCRIBE permission on
keyspace

Permission	Database resource	Description
ALTER	ALL ROLES ROLE <i>role_name</i>	ALTER ROLE on all roles or specified role.
CREATE	ALL ROLES ROLE <i>role_name</i>	CREATE ROLE with any name or only create a role with specific name, automatically is granted AUTHORIZE on the new role.
DESCRIBE	ALL ROLES ROLE <i>role_name</i>	LIST privilege on all roles or only roles granted to another, specified role.
DROP	ALL ROLES ROLE <i>role_name</i>	DROP ROLE on any role or permission to drop a specific role.

DSE: Object Hierarchy for GRANT|REVOKE



```
GRANT permission[, permission ...]  
ON 'filter_text' ROWS IN keyspace_name.table_name  
TO role_name;
```

Row-level access control (RLAC) is disabled by default.
To use RLAC, set `allow_row_level_security` parameter
to true in the `dse.yaml`.

DSE: Audit Secure Subsystem

- DataStax Enterprise (DSE) supports capturing database activity to a log file or table.
- The audit logger also captures queries and prepared statements submitted by DataStax drivers, which use CQL binary protocol.
- Security-
- Separation of duties
- Tuning, Capacity planning

```
CREATE KEYSPACE patientdata ...;  
CREATE ROLE security_admin;  
GRANT security_admin TO admin_guy;  
GRANT AUTHORIZE FOR  
SELECT, MODIFY, EXECUTE ON  
patientdata TO security_admin;
```

```
GRANT SELECT ON patientdata  
TO new_nurse;  
GRANT SELECT, MODIFY ON patientdata  
TO doctor_house;
```

